

Experiment – 04

- **Name of the Experiment:** Coefficient of friction for motion on horizontal plane.
- **Aim:**

To determine coefficient of friction for motion on horizontal plane

- **Introduction:**

When an external force is applied to a body to move it over the surface of another body, an opposing force comes into play along the common surface of contact of the two bodies. This opposing force is called friction or frictional resistance or frictional force.

Coefficient of friction, ratio of the frictional force resisting the motion of two surfaces in contact to the normal force pressing the two surfaces together. It is usually symbolized by the Greek letter mu (μ).

Mathematically,

$\mu = F/N$, where F is the frictional force and N is the normal force. Because both F and N are measured in units of force.

- **Apparatus Sketch:**

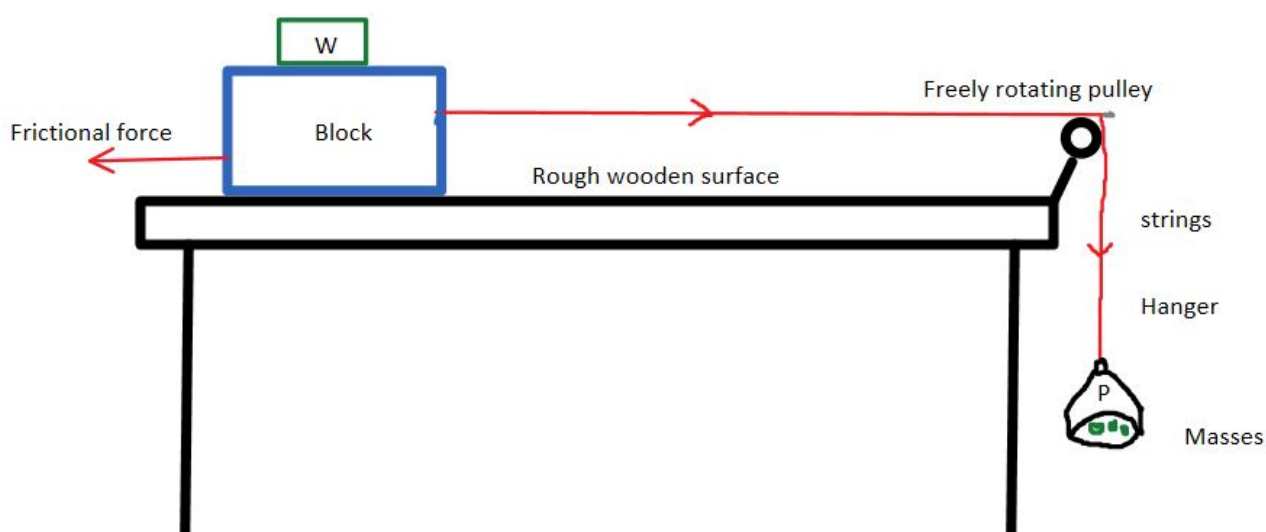


Fig. 4

- **Procedure:**

1. Use weighting scale to measure the mass of the frictional block and the mass hanger, and record their masses in the provided table.
2. Attach the one end of the cotton thread to the block and the other end to the mass hanger.
3. Place the block on the surface with thread over the pulley so that the hanger hangs free.

- Put a known mass on the block, and put smaller masses one by one into the mass hanger until the block starts to move. (Knock lightly against the surface after each smaller masses added). Record the readings in the results table.
- Repeat the procedure five times with five different masses on the block.

• **Observation Table:**

Mass of the Block(M_{Block})					
Mass of the Hanger (M_{Hanger})					
Sl. No.	Mass On Block	Mass On Hanger	$M_{Block} + M_{Mass\ on\ Block}$ (W)	$M_{Hanger} + M_{Mass\ on\ Hanger}$ (P)	$\mu = F/N = W/P$
1.					
2.					
3.					

• **Results:**

Co Efficient of friction $\mu =$

• **Precautions**

- Pulley must be frictionless.
- The weight should suspend freely without touching the table or other objects.
- Roller should roll smoothly, that is, without slipping.
- Weight, W should be decreased in small steps.